

Annotation Guide — ZW-BioBank v1.0

Label definitions, annotator workflow, review and adjudication · guide version v1.0.0

Purpose

This guide exists so that two annotators, working separately, apply the same label to the same record. Every definition below carries a positive example and a negative example. The negative example is the more important of the two: it names the specific mistake the label is most likely to attract.

Workflow

- 1. Assign.** The annotator records the label, their identifier, the timestamp and the version of this guide in `labels/annotations_v1.jsonl`. A label with no guide version is not traceable and is treated as unlabelled.
- 2. Review.** Every non-unknown `compound_class` assignment and every novelty call is independently double-annotated. All other labels are reviewed on a 10% sample.
- 3. Adjudicate.** Disagreements are recorded rather than silently resolved. A third reviewer adjudicates. The disagreement stays in the record, because a label two experts disputed is genuinely less certain than one they agreed on, and a model should be able to see that.
- 4. Version.** If a definition changes, this guide's version increments and affected records are re-reviewed. Labels are never silently redefined under an unchanged version.

The rule that matters most

Chemistry labels are never assigned from the traditional indication. If a holder describes an anti-parasitic use, that does not make the compound anti-parasitic. Assigning it so would launder the heritage record into a chemical claim and destroy the independence of the two streams — and the independence is the entire scientific value of joining them. The heritage record says where to look. The chemistry says what is there. Whether they agree is the finding, and it cannot be a finding if one was copied from the other.

Label definitions

LBL-01 · organism_novelty — genomic_sequences (category)

Allowed: known | divergent | putative_novel | unassessed

Novelty of the sequence against NCBI GenBank, from BLAST percent identity to the closest described organism. known >=98%; divergent 95-98%; putative_novel <95%; unassessed where no BLAST was run.

Correct: A 16S sequence at 91.2% identity to its nearest neighbour -> putative_novel

Incorrect: A sequence at 99.4% identity labelled putative_novel because the taxon is locally unfamiliar. Local unfamiliarity is not novelty.

Assigned by: automated (BLAST) then human review · **Review:** 10% human audit; any putative_novel call reviewed by a second bioinformatician

LBL-02 · predicted_function — genomic_sequences (string)

Allowed: free text, model-derived

Predicted protein or gene-cluster function from a structure or sequence model. Always recorded with the model and version that produced it.

Correct: 'non-ribosomal peptide synthetase (predicted)' from an antiSMASH cluster call

Incorrect: 'antibiotic' — a claim about activity, not a prediction about function. Activity requires wet-lab assay.

Assigned by: automated (model inference) · **Review:** spot-checked against cluster boundaries

LBL-03 · disease_target_relevance — genomic_sequences (float)

Allowed: 0.0 to 1.0

Model-estimated relevance of the sequence to one of the four declared disease targets. A prioritisation score for wet-lab triage, not a prediction of efficacy.

Correct: 0.81 for a BGC-dense actinomycete metagenome against AMR targets

Incorrect: Reporting 0.81 as an 81% chance the compound works. It is a ranking aid.

Assigned by: automated · **Review:** distribution reviewed per batch; not used as a sole basis for any claim

LBL-04 · annotation_confidence — genomic_sequences, metabolomic_profiles (float)

Allowed: 0.0 to 1.0

Confidence in the annotation itself, distinct from confidence in the underlying measurement.

Correct: 0.31 on an unmatched spectral feature

Incorrect: Leaving it blank because the annotation is a guess. A low score is information; a null is not.

Assigned by: automated · **Review:** audited alongside the label it qualifies

LBL-05 · compound_class — metabolomic_profiles (category)

Allowed: antimicrobial | anti-parasitic | anti-inflammatory | unknown

Putative activity class of the annotated compound, inferred from spectral library match and published activity of the matched compound. Never from the traditional indication alone.

Correct: Quercetin -> anti-inflammatory, from a level 1 library match and published activity

Incorrect: Assigning anti-parasitic because a holder described an anti-parasitic use. That would launder the heritage record into a chemical claim and destroy the independence of the two streams.

Assigned by: human annotator with pharmaceutical chemistry background · **Review:** double annotation on all non-unknown assignments; disagreement escalated to adjudication

LBL-06 · msi_annotation_level — metabolomic_profiles (integer)

Allowed: 1 | 2 | 3 | 4

Metabolomics Standards Initiative identification confidence. 1 confirmed against an authentic standard; 2 putative from spectral library; 3 putative class only; 4 unknown.

Correct: Chlorogenic acid at 0.98 library match with a matching InChIKey -> level 1

Incorrect: Calling a level 3 feature by a specific compound name in a report. Level 3 supports a class, not a name.

Assigned by: human annotator · **Review:** every level 1 claim independently verified before release

LBL-07 · compound_novelty — metabolomic_profiles (category)

Allowed: known | analogue | putative_novel | unassessed

Novelty of the compound against external natural-product databases.

Correct: An unmatched formula with no database hit -> putative_novel

Incorrect: Marking novel because it is absent from one library. Absence from a Western-biased library is the expected case here, not evidence.

Assigned by: automated then human review · **Review:** second reviewer on all novel calls

LBL-08 · heritage_link — genomic_sequences (string)

Allowed: tk_id or null

Denormalised foreign key to the heritage record that directed collection of the parent sample. Present so label joins do not require a three-table hop.

Correct: ZW-TK-2026-001 on a sequence from a knowledge-led sample

Incorrect: Populating it from geographic proximity. The link means 'this knowledge caused this collection', not 'these were near each other'.

Assigned by: derived from environmental_samples · **Review:** validator checks it matches the parent sample's heritage_use_ref

LBL-09 · translation_status — heritage_knowledge (category)

Allowed: pending | machine_draft | human_verified

Whether the English rendering of a verbatim vernacular record has been checked by a bilingual reviewer. The vernacular text is the scientific object; the translation is a derivative.

Correct: human_verified after a second Shona-speaking reviewer confirms the rendering

Incorrect: Releasing a machine_draft record. Records below human_verified are held out of every tier.

Assigned by: bilingual validator · **Review:** second bilingual reviewer; disagreements resolved with the holder where possible

LBL-10 · taxon_confidence — environmental_samples (category)

Allowed: field_tentative | herbarium_confirmed | sequence_confirmed

How a taxonomic identification was established. A label-noise indicator for any model trained on species labels.

Correct: A specimen matched to a National Herbarium accession -> herbarium_confirmed

Incorrect: Upgrading to herbarium_confirmed because the field team was confident.

Assigned by: field team, then National Herbarium, then sequencing · **Review:** field_tentative records down-weighted in training and flagged in any release